# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** According to the first law of reflection, the incident ray, the reflected ray and the normal at the point of incidence: 1

**(a)** are always equal in length **(b)** lie in the same plane
**(c)** are at right angles to each other **(d)** are parallel

**2.** The reflection from a rough surface in which parallel rays are reflected in different directions is: 1

**(a)** regular reflection **(b)** refraction
**(c)** total reflection **(d)** diffuse reflection

**3.** The relation between the focal length (f) and the radius of curvature (R) is: 1

**(a)** R = f / 2 **(b)** R = 2f **(c)** f = 2R **(d)** R = f

**4.** According to the second law of reflection: 1

**(a)** angle of incidence is double the angle of reflection
**(b)** angles are always 90 degrees
**(c)** angle of reflection is zero
**(d)** angle of incidence equals angle of reflection

**5.** Reflection from a smooth mirror is: 1

**(a)** regular reflection **(b)** diffuse reflection
**(c)** refraction **(d)** dispersion

**6.** The mirror used as a rear-view mirror in vehicles is: 1

**(a)** cylindrical **(b)** plane **(c)** convex **(d)** concave

**7.** A ray strikes a plane mirror at an angle of incidence of 35 degrees. The angle of reflection is: 1

**(a)** 70 degrees **(b)** 55 degrees **(c)** 35 degrees **(d)** 145 degrees

**8.** We can see a non-luminous object because it: 1

**(a)** reflects light into our eyes **(b)** is hot
**(c)** emits light **(d)** absorbs all light

**9.** A dentist uses a: 1

**(a)** plane mirror **(b)** concave mirror **(c)** convex mirror **(d)** no mirror

**10.** The angle of incidence is measured between the incident ray and the: 1

**(a)** mirror surface **(b)** reflected ray **(c)** normal **(d)** horizontal

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** State the laws of reflection. 2

**12.** What is lateral inversion? Why is 'AMBULANCE' written inverted on the front of the vehicle? 2

**13.** Differentiate between convex and concave lenses (one point each). 2

**14.** Differentiate between regular and diffuse reflection. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Draw in words the ray diagram of reflection at a plane mirror and mark i and r. 3

**16.** Differentiate between a concave and a convex mirror in three points. 3

**17.** Mention three uses of lenses. 3

OR

A ray of light makes 40 degrees with the surface of a plane mirror. Find the angle of incidence and the angle of reflection. 3

**18.** Name the parts of a spherical mirror: pole, centre of curvature, principal axis, focus. Define any two. 3

OR

Explain how a convex lens forms a real image of a distant object (such as the Sun) on a screen. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** State and explain the laws of reflection with a labelled description. Verify the laws using an activity. 5

OR

Describe the lens types, their action on light, the names of the principal focus and the focal length with uses. 5

**20.** Describe the characteristics of the image formed in a plane mirror. Two plane mirrors are inclined at 90 degrees. Find the number of images and explain. 5

OR

Describe an activity to find the approximate focal length of a convex lens using the Sun. 5

## END OF QUESTION PAPER — Total: 40 Marks

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